Online Resource 1 for "When can a prediction replace a physical trial? A production-referenced evaluation of design-stage decision rules in deep drawing"

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Abstract

This resource gives the specification of the rules and analyses added in the current version (Section 1), the further results that the main text cites as Tables S1–S20 (Sections 2–6) and four further figures (Figures S1–S4). Every number in the tables is written by `scripts/make_tables.py` from the result files of the archived release (Canseven 2026, Zenodo); the script and file behind each table are named in Section 7. Distances are in within-run floors of the characteristic's own geometry, a new family in the floor of the produced family unless stated otherwise; "<1" marks distances below one floor. M3n is named M4 in the code. A new process setting is interpolated (300 kN held out) or extrapolated (100 or 500 kN held out).

1 Specification of the rules and analyses added in this version

Trend GP (M7n) and autoregressive multi-fidelity model (M7).

Within one family, with x the force (100 kN) and the lubrication rank, the true 95th percentile of alternative i in pooled floors is modelled as

$$y_i = h(x_i)^\top \beta + f(x_i) + e_i, \qquad f \sim \mathrm{GP}\big(0, \sigma_f^2 k_\ell\big), \qquad e_i \sim N\big(0, s_i^2 + \sigma_n^2\big), \tag{S1}$$

with $h(x) = (1, \text{force}, \text{lubrication})$ (and the geometry indicator when both families calibrate), k_ℓ an anisotropic squared-exponential kernel, s_i^2 the block-bootstrap sampling variance of y_i (blocks of 50 parts) and σ_n^2 a between-run nugget. The coefficients β have a flat prior and are integrated out, so that the restricted likelihood is maximised and the predictive variance contains the uncertainty of the mean (universal kriging). The hyperparameters are maximum a posteriori under log-normal priors with medians $\ell_{\text{force}} = 200\,\text{kN}$, $\ell_{\text{lubrication}} = 1$ pattern step, $\sigma_f = 2$ floors and $\sigma_n = 1$ floor, and log-scale standard deviations 1, 1, 1.5 and 1, from six starts. The nugget is estimated and not bounded below by the variance between the lubrication series, which contains the lubrication effect that the mean and f represent. The 90 % interval is the predictive mean ± 1.645 predictive standard deviations, the latter including the nugget and the mean sampling variance of the calibration alternatives. M7 is the autoregressive model of Kennedy and O'Hagan (2000) with the nominal simulation $s_0(x)$ as the low-fidelity output: the same model with $h(x) = (1, s_0(x), \text{lubrication})$, so that the scale on s_0 is estimated by generalised least squares together with the GP discrepancy f. M1s is its special case without f.

Friction calibration with discrepancy (Mcd).

In the line of Kennedy and O'Hagan (2001), the 95th percentile of characteristic c is $y = \eta_c(x, \theta) + \delta_c(x) + e$, with η_c the simulation at the nominal thickness and friction θ, a uniform prior on the 11 friction values of the grid (0.05–0.15), and δ_c a GP with a constant mean and the priors of M7n. The posterior of θ is computed from the two draw-ins of the calibration alternatives, by the marginal likelihood of their discrepancy GP at each θ; the prediction for any characteristic is the posterior mixture over θ of the GP predictions, and its central 90 % interval is the verdict interval. Mc, the least-squares fit of one friction value per lubrication pattern without a discrepancy, is retained for comparison.

Relation-matched intervals (P+RM).

For a point predictor P (NN, M1sn, M1s, and the centres of M5n, M5, M3n, M7n and M7), the interval is $P \pm q$, with q the $\lceil 0.9(n+1) \rceil$-th smallest of P's n absolute errors in floors over the cases of the same evidence relation in the other family (sibling 63 cases, interpolated 21, extrapolated 42, new family 63, the other transfer). The family's own calibration set holds no case of its new-setting relation, which is why the errors are borrowed; the construction is split-conformal and covers at least 90 % if the two families' errors are exchangeable, which nothing guarantees.

Few-shot transfer.

$k = 1$–3 alternatives of the target family are produced, over every subset of size k, and its remaining $9 - k$ alternatives are judged in the target family's floor by: the nearest produced setting among the k (NN in family); the other family's 95th percentile at the same setting plus the median offset of the k from it (shifted record); a two-task GP of intrinsic coregionalisation type (Bonilla et al. 2008) with family intercepts, shared slopes, a free inter-family correlation $\rho = \tanh(\cdot)$ under a standard normal prior on

its argument, and otherwise the priors of M7n (ICM); and the same with the nominal simulation in the mean (ICM+sim).

Decision-time strata.

A new variant is classed by whether its two produced siblings differ resolvably from each other (effect-to-scatter ratio at least one between them), which a team knows before it decides. The strata of the first version, by whether the held-out alternative itself differs resolvably from its siblings, use its production and are kept only as a diagnostic (Table S3).

Guard bands out of sample, per family and per prior.

The band of step 5 is estimated per geometry and, out of sample, without the held-out alternative (within its family or pooled over both) and, for a new setting, without the held-out geometry–force cell; the share correct among decided verdicts and the trial rate are then computed on the held-out cases, with 95 % intervals from 2000 resamples of the alternatives within each geometry. The prior is the evaluation grid within $\pm h$ floors, $h = 3$, 5, 10, 20 (reference) and 50, every feasible distance and case weighted equally.

Costs and the trial as a rule.

The expected cost per verdict, Eq. (2) of the main text, is averaged over the reference prior for each rule with its 95 % band and with the band (0 to 40 floors in steps of 0.25) that minimises it, for $c_{\mathrm{T}} = 0.01$–0.5 and $c_{\mathrm{FR}} = 0.25$, 0.5 and 1 relative to c_{FA}. The trial of n parts is scored as a point rule that predicts the 95th percentile of the first $n = 20$ or 50 parts of the design's own run, a within-run lower bound of a trial's error.

Between-run sensitivity.

Every true 95th percentile is replaced by $q_{95} + \tau F z$, $z \sim N(0, 1)$ independently per alternative and characteristic, with $\tau = 0$, 0.5, 1, 1.5, 2 and 3 floors and, as a bound, per characteristic the variation between the series of one geometry and force ($\tau = 95\,\%$ quantile of their centre differences$/(1.96\sqrt{2})$); 200 draws per τ (50 for the bands), fits fixed.

Cross-fitting, design level and variants.

The rule with the smallest decisive (or safe) distance, and the guard band of step 6, are selected on the cases of one geometry and evaluated on the other (for a new family, on one transfer and the other). A design-level verdict accepts a design only if all seven characteristics meet: requirements lie δ floors above every truth (an adequate design) or δ floors below on one characteristic and above on the others (a failing design, every feasible choice of the failing characteristic weighted equally), both sides with equal weight. The variants refit the alternative-level rules with lower limits on the 5th percentile of the draw-ins, the wall angle and the depth (a change of sign), with truths and calibration from the first 20 or 50 parts of every run, and with the decided statistic $\bar{m} + 3.99\sigma$ in place of q_{95} (a performance index of 1.33 at the limit), and the

envelope of M2 and M5 over the 5th to 95th percentiles of the measured thickness and mapped friction instead of the whole grid.

2 Distances of all rules, their variants and strata

Table S1 gives every rule in every scope, including the pooled scopes, in which the other geometry's alternatives join the calibration set; M1n, M2n and M1sn have no family term there, so their pooled distances are not findings about the rules but about pooling two families without one. Table S2 gives the robustness and sensitivity variants and Table S3 the sibling strata. Fig. S1 shows the centres of the 18 alternatives, Fig. S2 the verdict shares of four further rules and Fig. S3 the calibration budget: the distances against the number of produced alternatives.

3 Intervals and their coverage

Table S4 gives the coverage of every interval rule with cluster-bootstrap intervals and its interval score, and Table S5 the coverage per characteristic and decision-time stratum. With a sibling the conformal margin of M6 costs about ten floors of decisiveness against M5 (17 against 7.2 floors; Table S1) and covers 93 %; for a new setting its margin, computed on sibling-type residuals of the calibration set, covers only 61 % extrapolating (Table S4).

4 Guard bands

Table S6 gives the step-6 rule pooled over the families, as in the first version; Tables S7–S10 the bands per family, out of sample, against the width of the prior and per characteristic; Fig. S4 the share correct among decided verdicts against the margin.

5 Costs, sensitivity, selection and design-level verdicts

Table S11 gives the cheapest rule per relation and costs, Table S12 the between-run sensitivity, Table S13 the cross-fitted selection and Table S14 the design-level verdicts.

6 Per characteristic, few-shot transfer, paired differences, simulation and measurement

Table S15 gives the distances per characteristic for the main rules, Table S16 the few-shot transfer and Table S17 the paired differences and the variance decomposition. Table S18 compares the simulation with the measurement per geometry and force, Table S19 gives the checks of the oil-film mapping, the cup depth and the scanner's lateral scale, and Table S20 the redundancy of the scans and the components of the floor.

7 Where the tables come from

S1: `decisions.py`, `learners.py`, `robustness.py` (`dec_resolution`, `learn_resolution`, `learn_source_floor`, `panel_source_floor`, `rob_decisions`); S2: `robustness.py`, `qualify.py`, `simcheck.py` (`rob_decisions`, `rob_refit`, `qual_variants`, `sim_envelope_variant`); S3: `qualify.py`, `panel.py` (`qual_strata`, `panel_sibling_strata`); S4, S5: `qual_coverage`; S6: `panel_step6`; S7–S10: `qual_guard_family`, `qual_guard_oos`, `qual_guard_prior_width`, `qual_guard_characteristic`; S11–S14: `qual_costs`, `qual_between_run`, `qual_crossfit`, `qual_design_level`; S15: `qual_per_characteristic`; S16: `learn_fewshot`; S17: `dec_paired`, `learn_paired`, `qual_resolvable_difference`, `qual_decomposition`; S18, S19: `simcheck.py` (`sim_validation`, `sim_mc_friction`, `sim_oil_mapping`, `sim_depth_offset`, `meas_tooling_reference`); S20: `measurement.py`, `panel.py`, `sensitivity.py` (`meas_scan`, `panel_floor_protocol`, `sens_drift`). Figures: `figs.py`.

Table S1 All rules: decisive/safe distances (floors) by scope, with the 95 % interval of the decisive distance

Rule	Sibling	Setting	Interp.	Extrap.	Fam., prod.	Fam., own	Pooled var.	Pooled set.
M0	108	108	109	106	104	108	108	108
interval	101–115	101–115	107–109	101–109	84–112	101–115	101–115	101–115
M1s	4.9	19	19	19	121	122	15	20
interval	4.7–6.3	14–21	19–20	15–19	119–121	121–123	14–19	17–22
M2	23/<1	18/8.1	18/2.7	18/9.9	41/26	51/16	53/<1	53/1.0
interval	22–23	17–18	15–18	17–19	39–42	50–52	45–57	45–57
M5	7.2/<1	19/6.8	19/2.8	19/8.9	35/31	38/28	11/<1	28/5.4
interval	6.6–8.4	18–20	18–20	18–20	31–39	35–39	9.6–12	25–32
M7n	6.8/<1	15/5.2	7.4/3.4	15/5.9	132/130	132/130	8.5/<1	14/2.8
interval	6.0–7.3	12–15	7.1–8.1	15–16	131–132	130–132	7.4–11	12–16
NN	3.4	8.5	4.7	9.5	130	131	3.4	8.5
interval	2.0–4.5	5.5–10	4.3–5.1	6.6–11	129–131	129–131	1.9–4.5	5.5–10
M1sn	3.5	8.2	4.7	8.9	131	130	77	111
interval	3.1–4.2	6.7–9.0	4.3–5.1	8.6–9.5	129–131	129–131	75–77	106–112
NN+RM	6.3/<1	20/<1	9.6/<1	20/<1	293/4.5	327/4.4	–	–
interval	5.6–6.7	15–20	9.4–9.8	18–20	293–294	322–331		
M1	15	24	11	25	33	35	27	28
interval	12–18	21–25	10–11	24–25	31–36	33–37	24–28	24–29
M1n	9.1	10	4.8	12	131	131	130	130
interval	5.5–10	9.0–12	4.1–5.3	9.8–13	129–131	129–131	128–130	129–131
M2n	19/<1	15/5.9	13/<1	16/9.1	133/128	132/129	325/<1	325/4.0
interval	15–21	15–16	12–13	15–18	132–133	131–133	321–331	321–331
M5n	7.2/<1	17/6.1	16/<1	18/8.6	133/128	132/130	9.8/<1	16/2.7
interval	6.4–7.5	16–18	15–16	16–19	131–133	130–133	8.1–11	12–16
M7	7.1/<1	28/16	22/16	28/17	140/104	144/109	9.6/<1	15/2.6
interval	6.1–8.2	22–53	22–23	23–54	139–141	144–146	8.3–11	14–19
Mc	27	39	14	40	43	38	32	33
interval	24–29	34–40	13–14	39–42	35–48	36–45	29–34	29–34
Mcd	7.2/<1	24/5.0	22/1.6	24/5.8	30/19	28/19	11/<1	29/3.6
interval	6.1–7.9	23–27	20–22	23–31	25–32	28–29	8.9–11	23–30
M0w	119/94	119/94	119/94	119/77	117/79	119/94	119/94	119/94
interval	118–119	118–119	118–119	116–120	115–119	118–119	118–119	118–119
M3	23/<1	31/16	29/13	33/17	40/17	49/21	25/<1	32/5.1
interval	17–26	25–36	24–31	25–40	36–45	45–53	24–28	30–35
M3n	11/<1	13/7.4	15/7.0	13/7.8	135/126	133/128	15/<1	15/4.9
interval	10–13	13–15	13–15	12–14	134–135	131–133	14–16	15–19
M6	17/<1	56/6.2	47/<1	56/8.1	35/28	41/25	–	–
interval					34–39			
M1sn+RM	7.1/<1	17/<1	9.6/<1	18/<1	293/4.3	327/4.2	–	–
interval	6.6–7.8	16–18	9.4–9.8	17–18	293–294	322–331		
M5n+RM	6.1/<1	23/<1	9.6/<1	24/<1	293/4.5	325/4.3	–	–
interval	5.2–7.2	22–24	9.4–9.8	22–24	293–294	322–331		
M7n+RM	6.5/<1	17/<1	11/<1	18/<1	293/5.1	327/4.9	–	–
interval	5.6–7.4	15–18	10–12	17–18	293–294	325–331		
M3n+RM	8.1/<1	21/<1	21/<1	20/<1	292/4.8	325/4.6	–	–
interval	7.1–9.9	20–21	21–21	17–20	291–293	321–331		
M5+RM	6.0/<1	26/1.2	21/<1	26/2.2	52/<1	81/<1	–	–
interval	5.2–6.8	23–26	21–22	26–26	51–53	80–81		
M1s+RM	11/<1	39/5.1	43/6.9	31/4.8	243/<1	325/<1	–	–
interval	11–12	31–43	38–43	30–40	243–243	321–331		
M7+RM	6.7/<1	39/5.1	45/6.4	31/<1	248/<1	332/<1	–	–
interval	6.0–7.7	33–45	39–46	30–40	247–248	327–336		

Fam., prod. and Fam., own: a new family in the produced family's floor and in its own. Pooled: the other geometry's alternatives added to the calibration set. Intervals over 1000 resamples of the held-out alternatives with the fits fixed.

Table S2 Decisive/safe distances (floors) under the robustness and sensitivity variants

	Sibling				New setting			
Variant	M1s	M2	M5	NN	M2	M5	NN	M1sn
Reference	4.9	23/<1	7.2/<1	3.4	18/8.1	19/6.8	8.5	8.2
Temperature-adjusted	5.0	23/<1	7.2/<1	3.4	17/7.6	19/6.8	9.4	8.9
Warm-up dropped	6.5	27/<1	9.0/<1	4.1	21/8.7	24/8.7	12	8.8
Floor between series	2.9	11/<1	1.3/<1	<1	9.9/2.6	11/2.3	2.7	4.5
Floor: batches of 25	4.6	21/<1	6.4/<1	2.9	16/6.9	17/6.3	7.7	7.3
Floor: batches of 100	5.3	24/<1	8.2/<1	3.5	19/8.5	21/7.9	8.9	8.9
Floor: 90 % quantile	6.2	31/<1	8.8/<1	4.0	21/10	23/8.6	10	10
Floor: 99 % quantile	3.4	14/<1	5.2/<1	2.4	11/4.4	13/4.4	5.7	6.1
Calibration: 50 parts	5.0	22/<1	6.8/<1	3.4	18/7.5	19/6.2	8.4	7.7
Conformance $p = 0.90$	4.9	23/<1	7.2/<1	3.3	18/7.5	18/6.4	8.4	8.1
Conformance $p = 0.99$	5.1	23/<1	7.5/<1	3.3	18/8.3	20/6.9	8.7	8.4
Interval level 0.80		23/<1	6.4/1.0		18/8.1	17/8.1		
Interval level 0.95		23/<1	8.0/<1		18/8.1	20/6.2		
Response surface	4.8	25/<1	7.2/<1	3.4	18/7.6	20/6.9	8.5	8.2
GP: sampling bound			6.7/1.4			18/7.4		
Reference, extrapolated*	4.9	23/<1	7.2/<1	3.4	18/9.9	19/8.9	9.5	8.9
Lower limits (5th percentile)*	3.6	62/<1	6.4/<1	3.0	62/30	65/27	8.3	6.7
Short runs: 20 parts*	4.8	22/<1	7.2/<1	3.0	18/7.6	19/7.6	8.8	7.3
Short runs: 50 parts*	4.8	22/<1	7.1/<1	3.2	18/7.7	19/7.7	9.0	7.8
Statistic $\bar{m} + 4\sigma$*	5.4	23/<1	8.0/<1	3.4	18/9.4	20/9.5	11	9.7
Floors without the held-out series*	4.9	22/<1	7.0/<1	3.3	18/9.4	19/9.2	9.6	8.6
Envelope over measured inputs*		35/1.3	24/1.3		34/7.7	40/7.1		
Refit, δ_{d}	4.9–7.3	22–23	4.0–11	2.3–4.6	16–18	18–20	6.9–9.2	7.3–10

Rules refitted for every variant. *: added in this version; their new-setting columns are the extrapolated setting, to be read against the extrapolated reference. Floors without the held-out series: every case in the floor of its geometry recomputed without its own alternative. Lower limits: draw-ins, wall angle and depth. Short runs: calibration and truths from the first n parts. Envelope over measured inputs: thickness and mapped friction between the 5th and 95th percentiles of the parts. Refit: 95 % interval over 200 draws of each geometry's lubrication patterns with replacement.

Table S3 Sibling strata: from decision-time information and, as a diagnostic, from the outcome

	Decision time: produced siblings				Outcome (diagnostic)		
Rule	not resolvable	p90	resolvable	p90	0	1	2
NN	1.60	1.54	4.40	4.36	0.85	2.45	4.85
M1sn	3.05	2.98	4.20	4.19	3.05	3.80	4.45
M1s	5.90	5.45	4.75	4.72	6.85	5.00	4.60
M2	22.65/<1	8.52	22.45/<1	11.13	22.45/<1	24.95/<1	14.75/1.4
M5	6.15/<1	1.85	8.90/<1	3.82	6.05/<1	6.10/<1	9.65/2.3
M7n	5.55/<1	1.73	8.55/<1	4.44	–	–	–
NN+RM	5.80/<1	1.54	6.45/1.6	4.36	–	–	–

Decision time: the two produced siblings differ resolvably from each other or not (75 and 51 cases on 17 and 16 alternatives). p90: 90th percentile of the absolute error of the centre (floors). Outcome: number of siblings that differ resolvably from the held-out alternative itself, which uses its production (58, 34 and 34 cases).

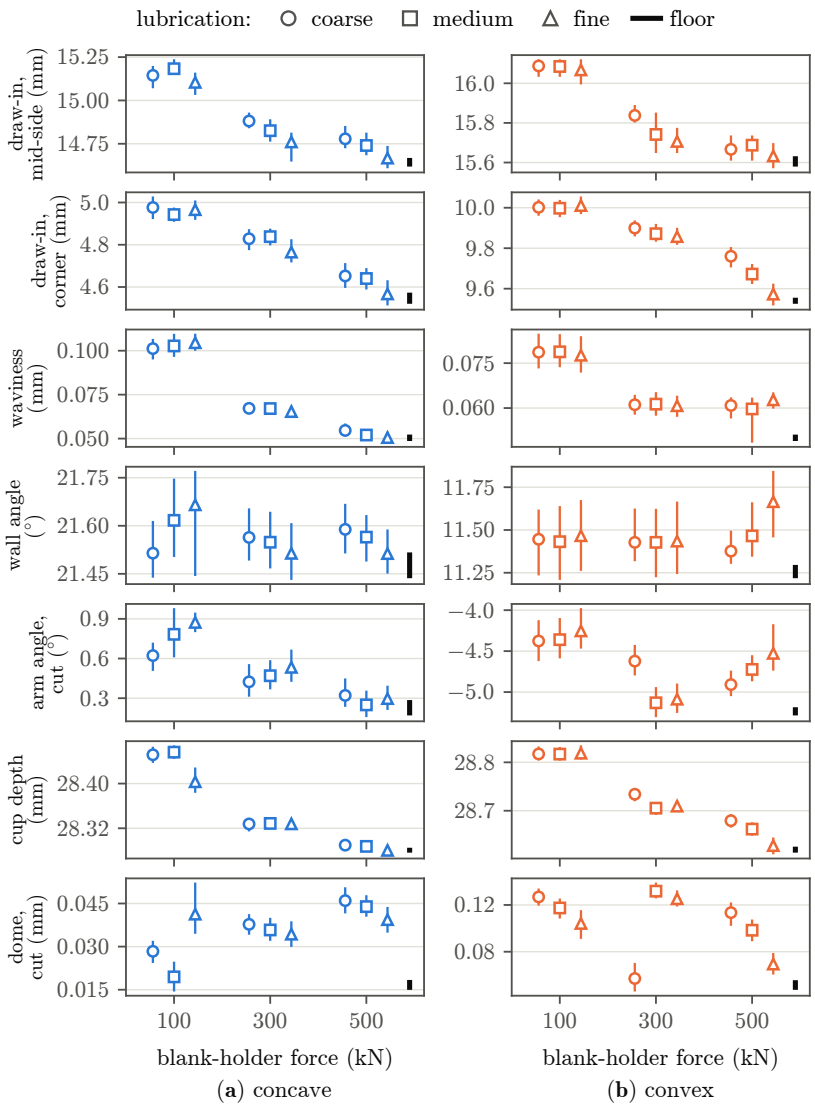


Fig. S1 Centres of the 18 alternatives (all 500 parts, with 95 % intervals over batches) per characteristic and geometry, and one floor (bar)

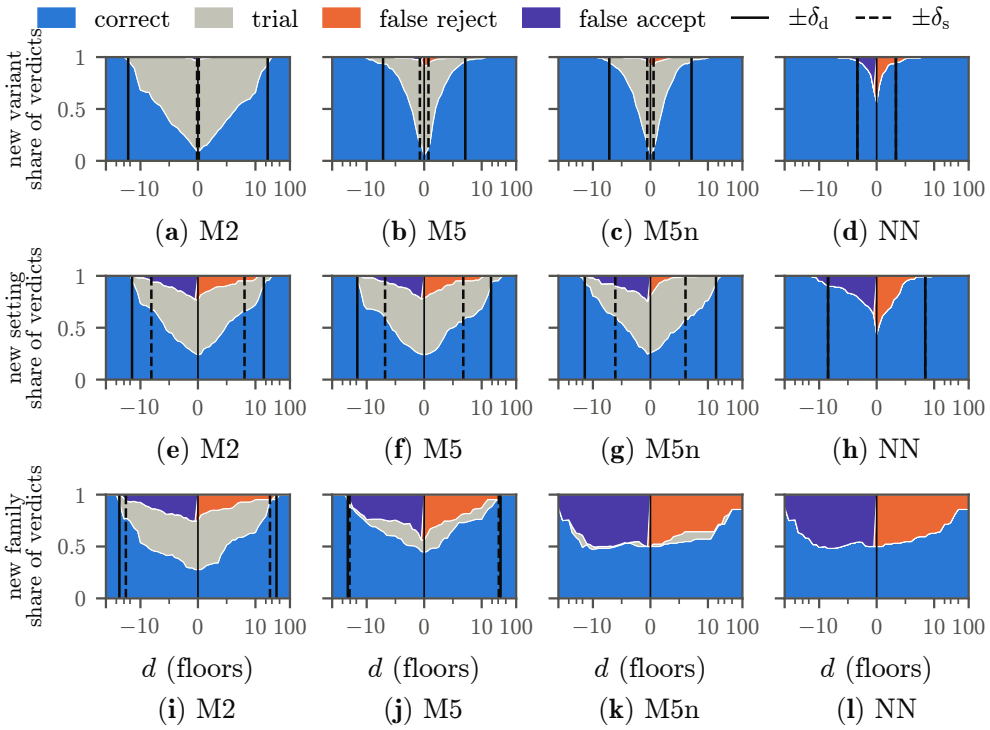


Fig. S2 Verdict shares of M2, M5, M5n and NN against the requirement distance for a new variant, a new setting (pooled) and a new family (produced family's floor)

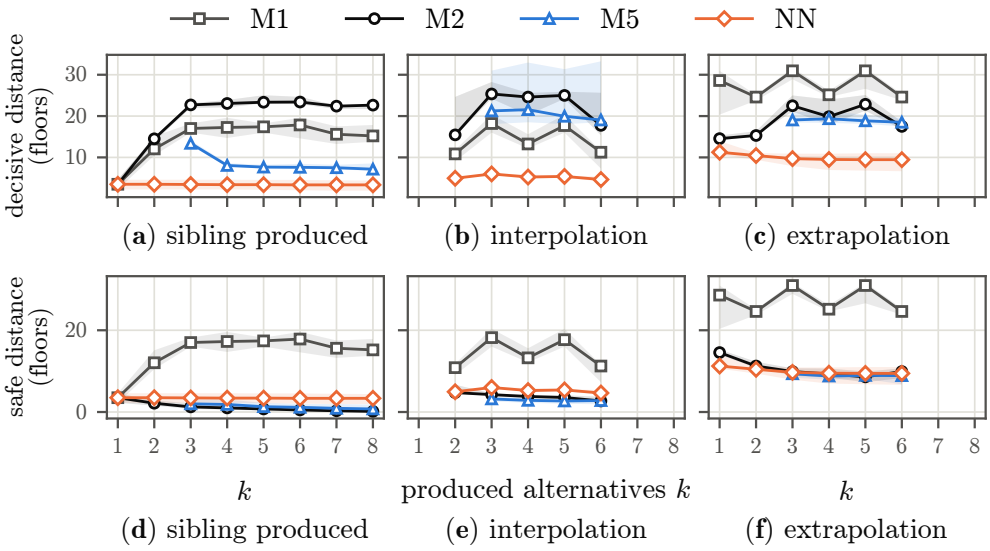


Fig. S3 Decisive and safe distances against the number k of produced alternatives, by the relation of the judged design to them; bands: 95 % over resamples

Table S4 Interval rules: coverage with cluster-bootstrap intervals and interval score by relation

		Sibling		Interpolated		Extrapolated		New family	
Rule	Level	Cov	IS	Cov	IS	Cov	IS	Cov	IS
M2	0.78	89 (83–94)	19	71	27	44	79	47	124
M2n	0.78	90 (82–96)	15	93	14	33	59	12	969
M3	0.78	96 (94–98)	21	43	98	43	120	40	134
M3n	0.78	89 (81–95)	11	36	65	48	48	7	989
M5	0.90 nominal	86 (75–94)	10	71	29	46	75	12	188
M5n	0.90 nominal	85 (76–93)	9	93	16	36	55	0	1024
M7	0.90 nominal	83 (73–92)	10	57	81	58	147	17	698
M7n	0.90 nominal	85 (75–94)	10	52	26	56	41	0	1043
Mcd	0.90 nominal	83 (72–91)	10	79	24	67	50	11	139
NN+RM	0.90	84 (74–93)	11	88	12	93	24	86	362
M1sn+RM	0.90	88 (81–94)	9	88	12	88	19	86	362
M7n+RM	0.90	91 (86–96)	11	90	14	89	19	86	363
M6	0.90	93 (87–98)	15	93	38	61	81	32	145
NN	point	–	24	–	42	–	81	–	1067
M1sn	point	–	32	–	42	–	77	–	1067
M1s	point	–	43	–	122	–	217	–	860

Level: guaranteed (largest residual, jackknife+; with a sibling) or nominal coverage. Cov: coverage (%), with a sibling with a 95 % interval over 2000 resamples of the alternatives within each geometry; for a new setting and a new family the alternatives of one held-out cell or family share their calibration set and are near-replicates, so with 2, 4 and 2 such units no useful interval can be given. IS: mean interval score (Gneiting and Raftery 2007) at $\alpha = 0.1$, floors; lower is better; for point rules $20\,|e|$. M6: conformal margin on standardised leave-one-out residuals of the calibration set, guaranteed only under exchangeability with them. A new family in the produced family's floor.

Table S5 Coverage (%) of the interval rules per characteristic and decision-time stratum

Relation	Rule	Mid-side	Corner	Wavin.	Wall	Arm	Depth	Dome	Not res.	Res.
sibling	M5	89	89	89	83	83	83	83	89	80
	interval	72–100	72–100	72–100	67–100	67–100	67–100	67–100	80–97	67–95
	M5n	89	83	89	72	89	89	83	88	80
	interval	72–100	67–100	72–100	50–89	72–100	72–100	67–100	79–96	67–93
	M7n	83	89	100	89	72	83	78	89	78
	interval	67–100	72–100	100–100	72–100	50–89	67–100	56–94	80–97	65–93
	M3	100	100	100	89	94	94	94	96	96
	interval	100–100	100–100	100–100	72–100	83–100	83–100	83–100	92–100	91–100
	NN+RM	100	89	100	89	72	78	61	95	69
	interval	100–100	72–100	100–100	78–100	56–89	61–94	39–78	87–100	52–88
interpolation	M5	100	100	50	50	100	67	33	–	–
	M5n	100	100	100	100	67	100	83	–	–
	M7n	17	100	0	100	67	33	50	–	–
	M3	0	67	67	67	50	17	33	–	–
	NN+RM	100	100	67	100	67	100	83	–	–
extrapolation	M5	0	50	50	50	58	50	67	–	–
	M5n	33	0	33	67	25	17	75	–	–
	M7n	33	92	0	67	92	17	92	–	–
	M3	17	17	50	42	67	25	83	–	–
	NN+RM	100	100	75	100	100	75	100	–	–

18 cases per characteristic with a sibling, 6 interpolated, 12 extrapolated. Not res., Res.: the produced siblings do not or do differ resolvably from each other (75 and 51 cases). Interval: 95 % over 2000 resamples of the alternatives within each geometry, given only with a sibling (see Table S4).

Table S6 The decision rule of step 6, pooled over the families: band, distances and trial shares

	Sibling				Interpolated				Extrapolated			
Rule	g	δ_d/δ_s	T_{10}	T_{20}	g	δ_d/δ_s	T_{10}	T_{20}	g	δ_d/δ_s	T_{10}	T_{20}
M1	none	–	100	100	8.75	20/2.1	44	4	none	–	100	100
M1s	2.0	6.9/2.9	1	0	none	–	100	100	none	–	100	100
M2	0	23/<1	33	8	0.75	19/2.0	31	4	none	–	100	100
M5	0	7.2/<1	1	0	0.5	20/2.3	40	4	none	–	100	100
M5n	0	7.2/<1	0	0	0	16/<1	25	0	6.75	25/1.9	48	15
NN	0.25	3.6/3.1	0	0	2.0	6.7/2.7	0	0	7.0	17/2.5	26	1
M3	0	23/<1	45	6	17.75	47/<1	94	69	16.75	51/<1	85	69
M3n	0	11/<1	9	0	6.0	21/1.0	55	6	6.5	19/1.3	47	3
M1sn	0.75	4.3/2.8	0	0	2.0	6.7/2.7	0	0	6.0	15/2.9	21	0

g: guard band (floors) under the reference prior; δ_d/δ_s: distances of the rule with its interval widened by g; T_{10}, T_{20}: share of verdicts sent to trial (%) at requirements 10 and 20 floors from the truth. None: no band up to 20 floors reaches 95 %; so also M0 and M0w in every relation and every rule for a new family.

Table S7 Guard bands estimated per family and moved between the families

			Concave			Convex		
Relation	Rule	Pooled g	g	Correct	Other g	g	Correct	Other g
sibling	NN	0.25	0	96	97	0.5	96	94
	M1sn	0.75	0.5	95	97	1.0	95	94
	M2	0	0	100	100	0	99	99
	M5	0	0	99	99	0	98	98
	M5n	0	0	99	99	0	99	99
	M7n	0	0	98	98	0	98	98
	Mcd	0	0	98	98	0	98	98
	NN+RM	0	0	100	100	0	98	98
interpolation	NN	2.0	1.5	95	97	2.5	95	93
	M1sn	2.0	1.5	95	97	2.5	95	93
	M2	0.75	0.75	95	95	0.5	95	96
	M5	0.5	1.75	95	91	0	97	99
	M5n	0	0	100	100	0	98	98
	M7n	0.75	0	96	99	1.75	95	92
	Mcd	0	0	100	100	1.25	96	92
	NN+RM	0	0	100	100	0	98	98
extrapolation	NN	7.0	9.5	95	92	5.75	95	100
	M1sn	6.0	5.5	95	97	6.5	95	93
	M2	none	none	–	84	6.25	95	–
	M5	none	none	–	83	5.0	95	–
	M5n	6.75	8.5	95	92	5.0	95	100
	M7n	3.25	2.75	95	98	4.75	95	93
	Mcd	6.25	8.0	95	92	4.5	95	99
	NN+RM	0	0.5	95	95	0	100	100
new family	NN	none	none	–	–	none	–	–
	M1sn	none	none	–	–	none	–	–
	M2	none	none	–	–	none	–	–
	M5	none	none	–	–	none	–	–
	M5n	none	none	–	–	none	–	–
	M7n	none	none	–	–	none	–	–
	Mcd	none	18.25	95	–	none	–	73
	NN+RM	none	none	–	59	none	–	–

Reference prior. Correct: share correct among decided verdicts (%) with the family's own band; Other g: with the band of the other family. For a new family, the family is the target of the transfer. None, –: every design to trial.

Table S8 Guard bands validated out of sample under the reference and the release-like prior

		In sample		LOAO, family		LOAO, pooled		Force cell out	
Prior, relation	Rule	g	Corr.	Corr.	T	Corr.	T	Corr.	T
reference, sibling	NN	0.25	96	95 (94–97)	2	95 (94–97)	2		
	M1sn	0.75	95	95 (94–97)	7	95 (94–97)	6		
	M2	0	99	99 (98–100)	54	99 (98–100)	54		
	M5	0	99	99 (98–100)	23	99 (97–100)	23		
	M7n	0	98	98 (97–100)	19	98 (97–100)	19		
	NN+RM	0	99	99 (97–99)	28	99 (97–99)	28		
interpolation	NN	2.0	95	95 (94–96)	17	95 (94–96)	17	95 (94–96)	18
	M1sn	2.0	95	95 (94–96)	17	95 (94–96)	17	95 (94–96)	18
	M2	0.75	96	95 (95–96)	52	95 (94–96)	52	95 (95–96)	52
	M5	0.5	95	96 (96–97)	56	95 (95–96)	55	95 (95–95)	57
	M7n	0.75	95	95 (95–96)	28	95 (94–96)	26	95 (94–96)	28
	NN+RM	0	99	99 (99–99)	42	99 (99–99)	42	99 (99–99)	42
extrapolation	NN	7.0	95	94 (92–97)	56	95 (92–98)	55	93 (90–97)	55
	M1sn	6.0	95	95 (94–96)	48	95 (94–96)	48	95 (94–96)	48
	M2	none	–	95 (93–98)	85	–	100	83 (79–87)	90
	M5	none	–	95 (92–99)	82	–	100	81 (77–85)	90
	M7n	3.25	95	95 (93–96)	58	95 (94–96)	57	95 (94–96)	58
	NN+RM	0	97	97 (95–100)	74	97 (95–100)	72	97 (95–100)	72
release-like, sibling	NN	2.0	95	96 (93–98)	41	95 (91–98)	35		
	M1sn	2.75	96	95 (93–97)	47	95 (92–97)	48		
	M2	0	97	97 (94–99)	76	97 (94–99)	75		
	M5	0	96	95 (91–100)	67	97 (93–100)	78		
	M7n	6.75	97	84 (83–100)	91	84 (82–100)	95		
	NN+RM	0.5	95	96 (92–100)	72	94 (89–98)	68		
interpolation	NN	4.25	96	95 (94–97)	66	95 (94–96)	68	93 (93–94)	67
	M1sn	4.25	96	95 (94–97)	66	95 (94–96)	68	93 (92–94)	67
	M2	2.75	97	96 (93–99)	83	95 (93–96)	84	94 (93–95)	85
	M5	2.75	95	95 (93–97)	83	94 (93–95)	86	88 (87–88)	84
	M7n	3.75	96	95 (93–98)	73	94 (91–98)	81	87 (85–89)	74
	NN+RM	0	95	96 (95–97)	78	95 (94–96)	76	95 (94–96)	77
extrapolation	NN	14.25	95	95 (90–98)	90	93 (89–100)	98	72 (68–78)	93
	M1sn	8.5	96	95 (92–98)	84	96 (93–98)	84	96 (92–98)	85
	M2	none	–	94 (89–100)	97	–	100	62 (61–66)	94
	M5	none	–	94 (90–100)	97	–	100	61 (58–66)	94
	M7n	6.75	96	94 (90–100)	89	96 (93–99)	91	94 (90–98)	91
	NN+RM	5.25	97	97 (94–100)	95	93 (89–100)	98	73 (69–84)	94

Corr.: share correct among decided verdicts (%), with a 95 % cluster interval out of sample; T: share sent to trial (%). LOAO: band estimated without the held-out alternative, within its family or pooled over both; force cell out: without the held-out geometry–force cell.

Table S9 Guard band (floors) and trial rate (%, in brackets) against the half-width of the prior

Relation	Rule	±3	±5	±10	±20	±50
sibling	NN	8.0 (99)	2.0 (37)	0.5 (5)	0.25 (2)	0 (0)
	M1sn	4.5 (89)	2.75 (50)	1.0 (10)	0.75 (7)	0.5 (4)
	M2	0.5 (85)	0 (75)	0 (60)	0 (54)	0 (48)
	M5	6.5 (99)	0 (45)	0 (26)	0 (23)	0 (20)
	M7n	7.25 (99)	6.75 (99)	0 (22)	0 (19)	0 (17)
	NN+RM	6.0 (100)	0.5 (65)	0 (31)	0 (28)	0 (24)
interpolation	NN	5.0 (86)	4.25 (68)	2.5 (25)	2.0 (18)	1.5 (11)
	M1sn	5.0 (86)	4.25 (68)	2.5 (25)	2.0 (18)	1.5 (11)
	M2	3.25 (89)	2.75 (84)	1.25 (62)	0.75 (53)	0 (42)
	M5	3.5 (91)	2.75 (86)	1.0 (62)	0.5 (54)	0 (45)
	M7n	4.75 (93)	3.75 (82)	1.25 (36)	0.75 (27)	0.25 (20)
	NN+RM	0.75 (88)	0 (75)	0 (48)	0 (42)	0 (37)
extrapolation	NN	14.75 (99)	14.25 (98)	8.25 (69)	7.0 (55)	5.75 (40)
	M1sn	9.0 (90)	8.5 (84)	6.75 (59)	6.0 (48)	5.25 (37)
	M2	none (100)	none (100)	none (100)	none (100)	18 (83)
	M5	none (100)	none (100)	none (100)	none (100)	17.75 (83)
	M7n	7.25 (94)	6.75 (91)	4.25 (68)	3.25 (56)	2.25 (44)
	NN+RM	5.5 (99)	5.25 (98)	0 (79)	0 (72)	0 (63)

Prior: the evaluation grid of requirement distances within the half-width, equal weights; ±20 is the reference prior and ±5 the release-like one. No rule qualifies for a new family.

Table S10 Share correct (%) per characteristic with the pooled band, and bands per class of characteristic

Relation	Rule	g	Mid-side	Corner	Wavin.	Wall	Arm	Depth	Dome	g_{P}	g_{G}	g_{C}
sibling	NN	0.25	97	96	99	97	94	96	89	0	0	1.75
	M1sn	0.75	98	97	93	98	95	93	92	0.5	0.5	1.5
	M2	0	100	99	99	99	100	99	98	0	0	0
	M5	0	100	99	100	99	99	98	94	0	0	0
	M7n	0	99	99	100	100	98	98	92	0	0	0
interpolation	NN	2.0	97	100	88	100	93	92	96	2.0	1.5	2.5
	M1sn	2.0	97	100	88	100	93	92	96	2.0	1.5	2.5
	M2	0.75	100	100	90	95	100	99	88	0	0.25	1.5
	M5	0.5	100	100	99	94	100	94	83	0	1.0	2.5
	M7n	0.75	97	100	89	100	92	96	93	0.75	0	2.75
extrapolation	NN	7.0	100	98	89	100	100	86	100	7.0	11.25	2.5
	M1sn	6.0	100	100	88	100	90	91	100	6.0	6.0	6.25
	M7n	3.25	100	100	83	100	100	94	100	4.5	1.25	0

Reference prior. g_{P}, g_{G}, g_{C}: bands estimated for the process indicators (draw-ins, waviness), the drawn geometry (wall angle, depth) and the cut part (arm, dome).

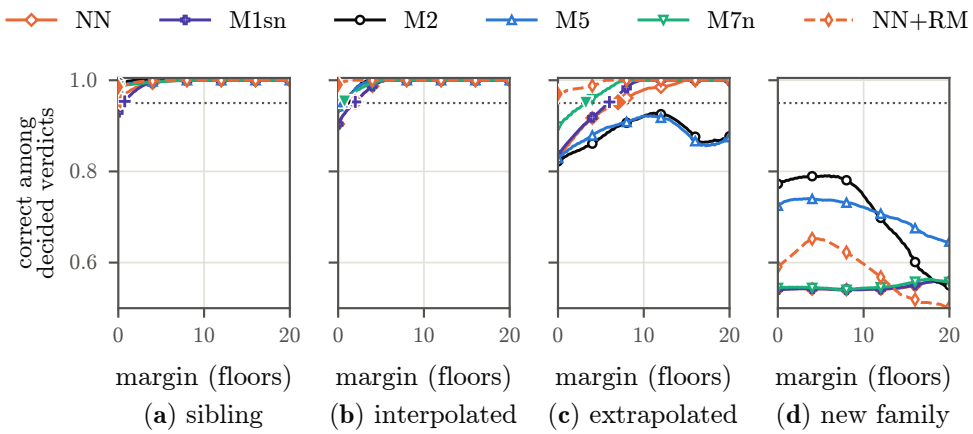


Fig. S4 Share correct among decided verdicts whose margin is at least the value on the axis, reference prior; markers: the guard band g where one qualifies

Table S11 Cheapest rule and its expected cost per verdict, with the cost-minimising band of NN

c_{FR}	$c_T = 0.02$	0.05	0.1	0.2	0.5
Sibling					
0.25	M1sn 0.008 (3.5)	M7n 0.015 (2.0)	NN 0.022 (1.0)	NN 0.027 (0.5)	NN 0.028 (0)
0.5	M1sn 0.009 (4.5)	M7n 0.018 (2.5)	NN 0.026 (1.5)	NN 0.033 (0.5)	NN 0.036 (0)
1	M1sn 0.009 (5.0)	M1sn 0.020 (3.0)	NN 0.031 (2.0)	NN 0.042 (1.0)	NN 0.051 (0)
Interpolation					
0.25	NN 0.009 (5.0)	NN 0.022 (4.0)	NN 0.033 (1.5)	NN 0.039 (0.25)	NN 0.040 (0)
0.5	NN 0.010 (5.0)	NN 0.022 (4.5)	NN 0.039 (3.0)	NN 0.054 (1.0)	NN 0.059 (0)
1	NN 0.010 (5.5)	NN 0.023 (5.0)	NN 0.043 (4.5)	NN 0.074 (3.0)	NN 0.096 (0)
Extrapolation					
0.25	M1sn 0.014 (14.75)	M1sn 0.034 (10)	M1sn 0.066 (8.25)	NN 0.108 (2.5)	NN 0.117 (0)
0.5	M1sn 0.014 (14.75)	M1sn 0.034 (10)	M1sn 0.067 (8.25)	M1sn 0.113 (3.0)	NN 0.135 (0)
1	M1sn 0.014 (14.75)	M1sn 0.035 (10)	M1sn 0.067 (8.25)	M1sn 0.119 (4.5)	M1sn 0.164 (0)
New family					
0.25	trial	trial	M3 0.098 (T)	M2 0.170 (T)	M5 0.215 (0)
0.5	trial	trial	trial	M2 0.183 (T)	M5 0.244 (0)
1	trial	trial	trial	M5 0.198 (T)	M5 0.303 (0)

Costs relative to a false accept; reference prior; each rule with its cost-minimising band (0 to 40 floors). Trial: sending every design to trial (cost c_T) is cheapest. In brackets: the cost-minimising band of NN (floors); T: NN is cheapest sending every design to trial.

Table S12 Decisive distance and guard band (floors, in brackets) with a between-run component added to the truths

Relation	Rule	$\tau = 0$	1	2	3	bound
sibling	NN	3.4 (0.25)	3.5 (0.5)	4.5 (1.5)	5.8 (2.9)	4.1 (1.0)
	M1sn	3.5 (0.75)	3.9 (1.0)	4.8 (2.0)	6.1 (3.25)	4.4 (1.5)
	M1s	4.9 (2.0)	5.6 (2.25)	6.2 (3.0)	7.1 (4.25)	6.1 (3.0)
	M2	23 (0)	23 (0)	23 (0)	23 (0.25)	23 (0)
	M5	7.2 (0)	7.5 (0)	8.3 (0)	9.3 (0.75)	8.3 (0)
	M7n	6.8 (0)	7.0 (0)	7.7 (0)	8.8 (1.0)	7.9 (0)
	record rule best	100	100	100	100	100
interpolation	NN	4.7 (2.0)	5.0 (2.25)	5.5 (3.0)	6.6 (4.0)	5.1 (2.75)
	M1sn	4.7 (2.0)	5.0 (2.25)	5.5 (3.0)	6.6 (4.0)	5.1 (2.75)
	M1s	19 (none)	20 (none)	21 (none)	21 (none)	19 (none)
	M2	18 (0.75)	18 (0.75)	19 (1.4)	20 (2.5)	19 (1.25)
	M5	19 (0.5)	20 (0.75)	21 (1.25)	22 (2.25)	20 (0.9)
	M7n	7.4 (0.75)	7.6 (1.0)	8.4 (1.5)	9.5 (2.6)	8.4 (1.25)
	record rule best	100	100	100	100	100
extrapolation	NN	9.5 (7.0)	9.5 (7.25)	9.5 (7.9)	10 (8.9)	9.7 (7.5)
	M1sn	8.9 (6.0)	8.8 (6.25)	9.2 (6.75)	9.9 (8.0)	9.0 (6.5)
	M1s	19 (none)	19 (none)	19 (none)	18 (none)	19 (none)
	M2	18 (none)	18 (none)	18 (none)	19 (none)	18 (none)
	M5	19 (none)	19 (none)	19 (none)	20 (none)	19 (none)
	M7n	15 (3.25)	15 (3.75)	16 (4.25)	16 (5.25)	16 (3.75)
	record rule best	100	100	100	100	100
new family	NN	130 (none)	131 (none)	132 (none)	133 (none)	131 (none)
	M1sn	131 (none)	131 (none)	132 (none)	133 (none)	131 (none)
	M1s	121 (none)	121 (none)	121 (none)	121 (none)	121 (none)
	M2	41 (none)	41 (none)	41 (none)	41 (none)	41 (none)
	M5	35 (none)	35 (none)	35 (none)	35 (none)	35 (none)
	M7n	132 (none)	132 (none)	133 (none)	134 (none)	132 (none)
	record rule best	0	0	0	0	0

Medians over 200 draws (bands over 50); τ: standard deviation of the component (floors); bound: each characteristic's variation between series, τ = 1.2–9.8/2.77. Record rule best: share of draws in which NN, M1sn or M1n is the most decisive rule (%).

Table S13 Rules and guard bands selected on one geometry and evaluated on the other

Relation	Criterion	Selected on	Rule	In sample	Other	g	Corr.	T
sibling	decisive, all rules	concave	NN	2.0	3.6			
sibling	decisive, all rules	convex	NN	3.6	2.0			
sibling	decisive, pre-specified	concave	M5	6.5	8.5			
sibling	decisive, pre-specified	convex	M5	8.5	6.5			
sibling	safe, all rules	concave	M1sn+RM	0	0.40			
sibling	safe, all rules	convex	M1s+RM	0	1.1			
sibling	safe, pre-specified	concave	M3	0	0			
sibling	safe, pre-specified	convex	M3	0	0			
sibling	step 6	concave	NN	2.0	3.6	0	94	0
sibling	step 6	convex	NN	4.1	2.5	0.5	97	4
interpolation	decisive, all rules	concave	M1n	3.6	5.8			
interpolation	decisive, all rules	convex	M1sn	5.3	4.2			
interpolation	decisive, pre-specified	concave	M1	12	6.8			
interpolation	decisive, pre-specified	convex	M1	6.8	12			
interpolation	safe, all rules	concave	M1s+RM	0	18			
interpolation	safe, all rules	convex	M3n+RM	0	1.8			
interpolation	safe, pre-specified	concave	M2	2.6	3.7			
interpolation	safe, pre-specified	convex	M5	1.9	3.1			
interpolation	step 6	concave	M1n	5.1	7.3	1.5	93	13
interpolation	step 6	convex	M1sn	7.8	6.8	2.5	97	22
extrapolation	decisive, all rules	concave	M1sn	8.7	8.9			
extrapolation	decisive, all rules	convex	NN	8.7	11			
extrapolation	decisive, pre-specified	concave	M3n	12	13			
extrapolation	decisive, pre-specified	convex	M3n	13	12			
extrapolation	safe, all rules	concave	M1sn+RM	0	0.25			
extrapolation	safe, all rules	convex	M1s+RM	0	11			
extrapolation	safe, pre-specified	concave	M3n	9.9	5.1			
extrapolation	safe, pre-specified	convex	M3n	5.1	9.9			
extrapolation	step 6	concave	M1sn	14	14	5.5	93	43
extrapolation	step 6	convex	NN	14	17	5.75	92	47
new family	decisive, all rules	concave	Mcd	29	33			
new family	decisive, all rules	convex	M1	33	32			
new family	decisive, pre-specified	concave	M1	32	33			
new family	decisive, pre-specified	convex	M1	33	32			
new family	safe, all rules	concave	M5+RM	0.05	0			
new family	safe, all rules	convex	M1s+RM	0	5.9			
new family	safe, pre-specified	concave	M3	18	17			
new family	safe, pre-specified	convex	M3	17	18			
new family	step 6	concave	Mcd	43	38	18.25	73	83
new family	step 6	convex	M5+RM	59	81	12	–	100

Selection by the smallest decisive or safe distance among all rules or the pre-specified ones, or by the smallest decisive distance of step 6; for a new family, selected on one transfer. Other: the distance on the other geometry. Corr., T: share correct and trial rate (%) there.

Table S14 Design-level verdicts: a design meets only if all seven characteristics meet

	Sibling		Interpolated		Extrapolated		New family	
Rule	Design	Marginal	Design	Marginal	Design	Marginal	Design	Marginal
M0	115/115	108/108	115/115	109/109	115/115	106/106	120/120	104/104
M1s	7.7/7.7	4.9/4.9	19/19	19/19	20/20	19/19	124/124	121/121
M2	27/1.4	23/0.15	26/2.8	18/2.7	27/11	18/9.9	43/34	41/26
M5	11/3.6	7.2/0.75	34/3.8	19/2.8	34/11	19/8.9	39/33	35/31
M7n	11/3.6	6.8/0.65	9.1/5.2	7.4/3.4	18/1.8	15/5.9	170/169	132/130
NN	5.1/5.1	3.4/3.4	5.3/5.3	4.7/4.7	9.9/9.9	9.5/9.5	170/170	130/130
M1sn	4.9/4.9	3.5/3.5	5.3/5.3	4.7/4.7	9.5/9.5	8.9/8.9	170/170	131/131
NN+RM	9.5/2.7	6.3/0.85	9.7/1.1	9.6/0.40	21/0	20/0	∞/46	293/4.5
M3	51/2.2	23/0	32/19	29/13	41/20	33/17	57/40	40/17

Decisive/safe distances (floors). Design: requirements on every characteristic δ floors above its truth (adequate design) or one characteristic δ floors below and the others above (failing design), both sides weighted equally. Marginal: per characteristic, as in the main text.

Table S15 Decisive/safe distances (floors) per characteristic

Relation	Rule	Mid-side	Corner	Wavin.	Wall	Arm	Depth	Dome
sibling	M0	38	115	15	28	34	115	32
	M1s	2.2	4.3	7.2	3.0	4.2	9.0	5.0
	M2	20/<1	6.1/<1	23/<1	9.3/<1	27/<1	28/<1	15/<1
	M5	3.4/<1	5.7/1.5	1.9/<1	4.8/1.8	7.2/1.2	5.4/<1	12/5.3
	M7n	3.4/<1	3.6/<1	2.5/<1	3.5/<1	6.8/<1	5.7/1.4	12/6.3
	NN	2.0	4.5	<1	3.4	2.8	3.3	7.0
	M1sn	2.6	3.5	3.6	3.0	4.6	3.8	6.0
	M3	26/<1	14/<1	21/<1	51/<1	25/<1	26/<1	36/<1
interpolation	M0	22	109	15	18	27	115	29
	M1s	2.8	1.3	19	<1	31	12	5.7
	M2	11/<1	7.1/<1	16/4.2	7.6/2.7	26/<1	14/<1	18/2.8
	M5	16/<1	5.1/<1	19/1.3	7.6/3.0	34/<1	17/2.2	17/3.8
	M7n	4.5/1.9	6.9/<1	6.1/3.8	3.9/<1	7.9/5.2	8.1/2.7	9.1/5.2
	NN	3.8	1.9	4.8	<1	5.9	4.4	5.3
	M1sn	3.8	1.9	4.8	<1	5.9	4.4	5.3
	M3	23/15	21/1.8	32/9.0	16/7.1	26/19	19/11	31/6.3
extrapolation	M0	38	115	8.6	28	35	109	32
	M1s	4.0	2.9	21	2.0	5.9	165	15
	M2	18/8.3	5.4/2.2	16/11	6.6/3.2	28/8.4	15/13	18/1.3
	M5	20/6.8	5.7/2.0	17/11	7.1/3.2	35/8.4	17/13	16/2.5
	M7n	7.9/3.6	12/<1	11/7.5	7.6/<1	19/<1	15/7.7	16/<1
	NN	6.5	6.7	11	1.9	5.4	15	4.3
	M1sn	5.7	2.8	9.3	1.8	9.5	9.9	3.0
	M3	34/19	20/11	25/16	23/8.1	44/20	22/13	39/3.0
new family	M0	40	113	26	29	40	120	33
	M1s	15	107	18	125	43	29	23
	M2	20/1.9	11/3.4	32/9.8	10/<1	46/34	25/5.9	40/36
	M5	5.5/2.1	9.9/3.9	22/21	7.5/3.1	39/32	20/15	40/37
	M7n	21/18	170/169	12/11	131/129	48/44	48/44	33/26
	NN	20	170	12	130	46	46	28
	M1sn	20	170	13	130	47	45	28
	M3	17/3.2	15/6.6	32/17	31/5.3	58/40	31/15	40/17

In the floor of the characteristic's geometry; a new family in the produced family's floor. 18 cases per characteristic with a sibling and for a new family, 6 interpolated, 12 extrapolated.

Table S16 Few-shot transfer to a new family with k produced alternatives of the target family

k	Relation	Cases	NN in family	Shifted record	ICM	ICM+sim	Cov ICM
1	all	1008	11	8.9	17/1.6	27/1.2	85
	sibling	252	3.5	4.8	13/0	13/0	92
	extrapolation	756	11	9.6	21/2.5	28/2.6	83
2	all	3528	7.7	7.8	14/1.3	19/1.3	83
	sibling	1638	3.5	5.9	12/0.10	13/0.05	89
	interpolation	378	5.0	5.8	13/1.3	15/3.8	78
	extrapolation	1512	10	9.5	16/3.1	24/3.4	78
3	all	7056	5.9	8.3	13/1.4	15/1.3	80
	sibling	4536	3.5	7.9	11/0.50	12/0.30	83
	interpolation	756	6.0	8.2	13/1.9	15/4.1	73
	extrapolation	1764	9.7	9.6	16/4.0	21/3.7	75

Decisive (and safe) distances in the target family's floor over every subset of k produced alternatives; relation of the judged alternative to the k. NN in family: nearest produced setting among the k; shifted record: the other family's q_p at the same setting plus the median offset of the k; ICM: two-task GP with family intercepts and a free inter-family correlation; ICM+sim: with the nominal simulation in the mean. Cov: coverage of the ICM interval (%).

Table S17 Paired differences of the distances between rules, their resolution and the decomposition

Pair $a - b$		Sibling	Interpolated	Extrapolated
M5−NN	δ_d	3.8 (2.8, 5.2)	14.4 (13.5, 15.2)	9.1 (6.9, 12.3)
	δ_s	−2.6 (−3.5, −1.6)	−1.9 (−2.6, −1.3)	−0.5 (−2.5, 2.9)
M5n−NN	δ_d	3.8 (2.6, 5.1)	10.9 (10.3, 11.3)	8.7 (5.6, 11.6)
	δ_s	−2.8 (−3.5, −1.7)	−4.7 (−5.1, −4.2)	−0.8 (−2.4, −0.5)
M2−NN	δ_d	19.3 (17.2, 20.6)	13.1 (10.4, 13.5)	8.1 (5.9, 11.4)
	δ_s	−3.2 (−4.4, −1.9)	−1.9 (−2.5, −1.5)	0.5 (−2.0, 3.4)
M1sn−NN	δ_d	0.1 (−0.7, 1.4)	0.0 (0.0, 0.0)	−0.6 (−1.9, 2.2)
	δ_s	0.1 (−0.7, 1.4)	0.0 (0.0, 0.0)	−0.6 (−1.9, 2.2)
M1s−M1sn	δ_d	1.3 (0.7, 3.0)	14.6 (14.4, 15.1)	10.3 (5.6, 10.5)
	δ_s	1.3 (0.7, 3.0)	14.6 (14.4, 15.1)	10.3 (5.6, 10.5)
M5−M5n	δ_d	0.0 (−0.4, 1.1)	3.5 (2.9, 4.1)	0.4 (−0.6, 2.2)
	δ_s	0.2 (−0.4, 0.6)	2.8 (2.2, 3.0)	0.3 (−1.8, 4.6)
M2−M2n	δ_d	3.9 (0.4, 7.3)	4.4 (2.1, 6.0)	2.0 (−0.2, 3.0)
	δ_s	0.1 (−0.5, 0.6)	2.7 (2.5, 2.8)	0.8 (−1.1, 4.6)
M3−M3n	δ_d	11.8 (4.4, 15.7)	14.3 (10.2, 16.2)	19.8 (11.6, 27.7)
	δ_s	−0.1 (−0.9, 0.1)	6.3 (2.9, 8.3)	9.2 (7.1, 12.2)
M7n−NN	δ_d	3.5 (2.5, 4.8)	2.8 (2.3, 3.5)	5.7 (4.2, 8.3)
	δ_s	−2.7 (−3.8, −1.5)	−1.3 (−1.7, −1.0)	−3.6 (−4.5, −1.1)
M7n−M5n	δ_d	−0.4 (−0.9, 0.5)	−8.2 (−8.6, −7.3)	−3.0 (−3.6, −1.0)
	δ_s	0.1 (−0.8, 0.8)	3.4 (2.8, 4.0)	−2.8 (−3.2, 0.7)
M7n−M1sn	δ_d	3.3 (2.6, 3.8)	2.8 (2.3, 3.5)	6.3 (5.7, 7.0)
	δ_s	−2.9 (−3.5, −2.3)	−1.3 (−1.7, −1.0)	−3.0 (−3.9, −1.6)
M7−M7n	δ_d	0.3 (−0.5, 1.1)	14.7 (13.8, 15.3)	13.2 (7.9, 39.2)
	δ_s	0.1 (−0.2, 0.5)	12.4 (6.9, 13.3)	10.7 (3.8, 11.6)
NN+RM−M5	δ_d	−0.9 (−1.9, −0.3)	−9.5 (−10.1, −8.9)	1.6 (−1.0, 2.6)
	δ_s	0.1 (−0.6, 1.1)	−2.4 (−3.0, −1.6)	−8.9 (−9.8, −6.8)
NN+RM−M7n	δ_d	−0.5 (−1.1, 0.1)	2.2 (1.4, 2.6)	5.0 (2.9, 5.5)
	δ_s	0.2 (−0.7, 1.4)	−3.0 (−3.2, −2.8)	−5.8 (−7.3, −4.8)
Mcd−Mc	δ_d	−19.5 (−21.8, −16.9)	7.9 (6.1, 8.7)	−15.4 (−18.6, −7.8)
	δ_s	−26.2 (−28.1, −22.7)	−12.4 (−12.9, −11.5)	−33.9 (−37.3, −30.0)
Median half-width of the paired 95 % intervals (floors)				
point rules		2.11 (1.02–2.93)	0.39 (0.00–0.77)	2.24 (1.32–3.93)
all pairs		2.11 (0.75–5.65)	0.80 (0.00–3.70)	2.44 (0.85–8.65)
Share of the variance of the distances of the pre-specified rules M1–M5				
all relations	δ_d	relation 58 %, rule 10 %, interaction 32 %		
within the family	δ_d	relation 16 %, rule 60 %, interaction 24 %		
all relations	δ_s	relation 71 %, rule 11 %, interaction 18 %		
within the family	δ_s	relation 54 %, rule 35 %, interaction 11 %		

Difference of rule a minus rule b (floors) with the 95 % interval over 1000 common resamples of the held-out alternatives, fits fixed; negative: a closer. Descriptive and not adjusted for the number of comparisons. Decomposition: two-way analysis of the distances of M1–M5 over relations and rules, in the produced family's floor.

Table S18 The simulation dataset against the measurement per geometry and force

Characteristic	Geometry	kN	Measured	$\mu = 0.10$	Range over μ	Inside	Closest μ	Residual
Draw-in, mid-side	concave	100	15.14	16.98	16.63–17.34	no	0.15	30.0
	concave	300	14.82	15.66	14.83–16.67	no	0.15	0.2
	concave	500	14.73	14.85	13.59–16.14	yes	0.11	−2.4
	convex	100	16.08	18.09	17.81–18.41	no	0.15	33.1
	convex	300	15.76	16.91	16.17–17.71	no	0.15	7.7
	convex	500	15.66	16.05	14.78–17.23	yes	0.12	−1.7
Draw-in, corner	concave	100	4.962	8.385	8.240–8.525	no	0.15	62.2
	concave	300	4.811	7.971	7.517–8.328	no	0.15	51.4
	concave	500	4.620	7.210	5.876–8.046	no	0.15	23.8
	convex	100	10.00	13.48	13.31–13.65	no	0.15	110.5
	convex	300	9.876	13.14	12.76–13.47	no	0.15	96.3
	convex	500	9.668	12.58	11.38–13.27	no	0.15	57.1
Flange waviness	concave	100	0.1028	0.08211	0.05973–0.09609	no	0.05	−1.8
	concave	300	0.06655	0.01440	0.01300–0.05128	no	0.05	−4.2
	concave	500	0.05241	0.03940	0.01075–0.07766	yes	0.12	0.9
	convex	100	0.07836	0.06604	0.05264–0.06648	no	0.14	−5.6
	convex	300	0.06106	0.05964	0.05690–0.08995	yes	0.10	−0.7
	convex	500	0.06110	0.07239	0.05734–0.1112	yes	0.07	1.4
Wall angle	concave	100	21.60	20.92	19.77–21.10	no	0.07	−6.2
	concave	300	21.54	20.32	19.44–20.61	no	0.11	−11.5
	concave	500	21.56	19.42	19.33–20.92	no	0.15	−8.0
	convex	100	11.45	10.74	10.01–10.91	no	0.11	−6.9
	convex	300	11.43	10.30	10.30–10.85	no	0.05	−7.5
	convex	500	11.50	11.13	10.31–11.20	no	0.14	−3.9
Arm angle, cut	concave	100	0.7594	−0.4379	−1.103 to 0.1517	no	0.05	−5.2
	concave	300	0.4757	−3.702	−4.363 to −1.142	no	0.05	−13.9
	concave	500	0.2892	−4.349	−4.451 to −2.928	no	0.05	−27.7
	convex	100	−4.330	−4.443	−4.687 to −3.104	yes	0.11	0.4
	convex	300	−4.947	−5.503	−5.503 to −4.246	yes	0.06	0.0
	convex	500	−4.720	−4.296	−5.586 to −2.301	yes	0.08	0.0
Cup depth	concave	100	28.44	29.35	29.35–29.38	no	0.10	104.3
	concave	300	28.33	29.34	29.31–29.36	no	0.06	112.1
	concave	500	28.29	29.24	29.00–29.33	no	0.15	82.3
	convex	100	28.82	29.88	29.81–29.88	no	0.05	80.8
	convex	300	28.72	29.64	29.47–29.80	no	0.15	61.5
	convex	500	28.66	29.37	29.12–29.64	no	0.15	37.9
Bottom dome, cut	concave	100	0.02971	0.1336	0.01244–0.1336	yes	0.15	0.2
	concave	300	0.03595	0.1387	0.03240–0.1753	yes	0.15	−1.0
	concave	500	0.04309	0.08286	−0.004812 to 0.1545	yes	0.08	2.1
	convex	100	0.1162	0.04951	−0.002123 to 0.06661	no	0.12	−6.0
	convex	300	0.1047	0.01768	0.001869–0.08873	no	0.12	−1.9
	convex	500	0.09375	0.02753	−0.0008228 to 0.08175	no	0.11	−1.5

Values in the unit of the characteristic (signed), simulation at 0.99 mm. $\mu = 0.10$: the nominal simulation; range over μ: over the friction grid 0.05–0.15. Residual: simulation at the closest grid friction minus the measurement (floors). Mc's friction fits end at the upper edge of the grid in 72 of 72 within-family calibrations.

Table S19 Checks of the simulation's oil-film mapping, the cup-depth offset and the scanner's lateral scale

Check	Case	a	b	c	d
oil film	concave, 100 kN, draw-in, mid-side	0.89	−0.008	−0.17	−0.37 to +2.00
oil film	concave, 100 kN, draw-in, corner	0.36	0.035	0.15	0.11–0.49
oil film	concave, 300 kN, draw-in, mid-side	2.30	0.091	0.36	0.25–0.60
oil film	concave, 300 kN, draw-in, corner	1.00	0.009	0.19	−0.04 to +0.67
oil film	concave, 500 kN, draw-in, mid-side	3.13	0.052	0.27	0.19–0.34
oil film	concave, 500 kN, draw-in, corner	2.69	0.050	0.20	0.06–0.34
oil film	convex, 100 kN, draw-in, mid-side	0.77	0.029	0.05	0.02–0.07
oil film	convex, 100 kN, draw-in, corner	0.43	−0.018	−0.02	−0.05 to +0.03
oil film	convex, 300 kN, draw-in, mid-side	1.93	0.069	0.39	0.35–0.53
oil film	convex, 300 kN, draw-in, corner	0.89	0.012	0.12	0.10–0.22
oil film	convex, 500 kN, draw-in, mid-side	2.99	0.023	0.07	−0.08 to +0.23
oil film	convex, 500 kN, draw-in, corner	2.29	0.010	0.39	0.35–0.42
Cup depth, flange rise and wall position at 15 mm depth (mm, simulation at nominal)					
depth	concave, 100 kN	0.90	0.30 / 0.24	2.05	0.008
depth	concave, 300 kN	1.01	0.17 / −0.09	2.20	0.020
depth	concave, 500 kN	0.96	0.16 / −0.29	2.02	0.039
depth	convex, 100 kN	1.06	0.15 / 0.09	2.16	0.006
depth	convex, 300 kN	0.92	0.12 / 0.03	2.08	0.053
depth	convex, 500 kN	0.70	0.14 / 0.05	2.06	0.006
A dimension fixed by the tooling: cup width at 15 mm depth					
tooling	concave	−0.27	0.062	0.099	0.049
tooling	convex	0.27	0.067	0.104	0.045

Oil film (mm of draw-in per g/m^2): a, implied by the dataset's mapping from oil film to friction and the simulated friction sensitivity; b, slope within the run; c, median and d, range of the slopes between the lubrication patterns. Depth: a, simulated minus measured cup depth; b, flange rise towards the edge, measured/simulated; c, wall position at 15 mm depth, measured minus simulated (radius); d, standard deviation of the press position at bottom dead centre within the series. Tooling: a, correlation of the batch centres of flange width and cup width; b, c, floors of the cup width and of the flange width (mm); d, bound of a lateral scale drift (%).

Table S20 Redundancy of the scans per series, and the components of the floor

Series	$W_y - W_x$	σ_y/σ_x	r_{xy}	Diagonals	N−EW	r_{EW}
concave/100/coarse	2.49	5.9	0.01	1.84	0.12	0.09
concave/100/fine	3.47	3.8	0.20	2.06	0.39	0.27
concave/100/medium	2.69	6.1	-0.12	1.94	0.15	0.08
concave/300/coarse	2.82	4.5	-0.04	1.79	0.12	0.16
concave/300/fine	2.48	4.4	-0.14	1.81	0.20	0.05
concave/300/medium	2.70	5.8	0.06	1.75	0.12	0.17
concave/500/coarse	3.29	4.7	0.00	1.74	0.13	0.04
concave/500/fine	2.86	5.1	-0.01	1.64	-0.05	-0.10
concave/500/medium	2.94	3.7	0.01	1.67	0.09	0.01
convex/100/coarse	3.46	17.7	0.05	1.94	-0.92	-0.66
convex/100/fine	2.29	9.5	-0.21	1.96	-0.96	-0.57
convex/100/medium	2.98	16.2	0.00	1.96	-0.92	-0.71
convex/300/coarse	2.33	7.7	0.09	2.17	-1.23	-0.55
convex/300/fine	2.44	7.9	-0.02	1.89	-0.98	-0.53
convex/300/medium	2.43	5.3	0.01	1.87	-0.92	-0.54
convex/500/coarse	2.32	5.4	0.02	2.16	-1.23	-0.60
convex/500/fine	2.40	6.1	-0.00	1.96	-1.64	-0.62
convex/500/medium	2.64	8.1	0.02	2.13	-1.33	-0.49
Components of the floor, concave/convex						
Draw-in, mid-side	σ_{b}/F 0.32/0.34	σ_{w}/F 0.91/0.66	drift 3.3			
Draw-in, corner	σ_{b}/F 0.31/0.35	σ_{w}/F 0.65/0.77	drift 4.5			
Flange waviness	σ_{b}/F 0.33/0.30	σ_{w}/F 0.55/1.31	drift 2.5			
Wall angle	σ_{b}/F 0.32/0.26	σ_{w}/F 0.68/1.39	drift 2.1			
Arm angle, cut	σ_{b}/F 0.36/0.31	σ_{w}/F 0.46/1.22	drift 2.6			
Cup depth	σ_{b}/F 0.31/0.34	σ_{w}/F 0.65/0.52	drift 3.9			
Bottom dome, cut	σ_{b}/F 0.33/0.36	σ_{w}/F 0.76/0.55	drift 3.9			

$W_y - W_x$: flange width in the direction of motion minus that along the laser line (mm); σ_y/σ_x: ratio of their part-to-part standard deviations; r_{xy}: their correlation; diagonals: difference of the two flange diagonals (mm); N−EW: north wall angle minus the mean of east and west (°); r_{EW}: correlation of east and west. Components: σ_{b} between batch means, σ_{w} within batches, drift: floor over that of permuted series.